

From the macroscopic to the microscopic

Particle model and kinetic theory

Physics · 3A · 6 October 2026

A puddle at 20 °C

(slide 2)

Before moving on. A puddle at 20 °C dries up. Water boils at 100 °C. Why does it dry? A: the whole puddle slowly heats up to 100 °C, then evaporates. B: a few water molecules escape from the surface, even when the water is cold. C: sunlight breaks the water molecules apart.

Before moving on. A drop of ink falls into a glass of cold water and another into a glass of hot water, at the same moment. Which one spreads faster?

We keep both questions open: we close them at slide 17.

We have never asked what matter is made of

(slide 4)

A drop of ink spreads in still water. A smell crosses the room. A gas fills any container it is put in.

Before moving on. What do these three things have in common? And how could we prove that particles exist, if they are too small to see?

What they have in common is *movement*: the ink moves without anyone stirring. The idea to test: matter is made of tiny particles that are always moving. But can we see them?

Brownian motion

(slide 5)

Before moving on. Under a microscope, a visible particle of smoke or pollen jiggles for no apparent reason. What is pushing it? A current? Something alive?

Brownian motion is the random, zigzag movement of visible particles, such as smoke or pollen, in a fluid. **Cause:** collisions with fast-moving molecules that are too small to see. **Evidence** that the molecules exist, and that they move.



Figure 1: Brownian motion. Left: a large particle hit by many small, fast molecules that are too small to see; at each instant more arrive from one side than from the other. Right: the zigzag path of the visible particle (schematic, not to scale).

The large particle is hit by many small, fast molecules, and at each instant more arrive from one side than from the other: hence the jiggle and the zigzag path. The cause is invisible by definition, which is why we show it. Robert Brown observed pollen grains in 1827; the explanation came from Einstein (1905) and the experiments of Perrin. The 0653 asks for the description and the cause, not the mathematics.

Solids, liquids and gases: complete the table

(slide 6)

Before moving on. Complete the table for solid, liquid and gas, choosing from the word bank. Word bank: fixed regular pattern, close together, far apart, random, vibrate, slide past each other, fast, strong, weaker, almost none, high, low, yes, no. The rows are: arrangement, movement, forces between particles, density, easily compressed?

	Solid	Liquid	Gas
Arrangement	fixed, regular pattern	close together, irregular	far apart, random
Movement	vibrate about fixed positions	slide past each other	fast, random motion
Forces between particles	strong	weaker	almost none
Density	high	high	low
Easily compressed?	no	hardly	yes

The reasons matter, not only the words. Why is a gas easily compressed? Because the particles are far apart: there is empty space.

The same particles, three states

(slide 7)

The table from the previous slide, now alive, in a two-dimensional model with attractive forces between the particles (not a real substance; the temperature is in arbitrary units).

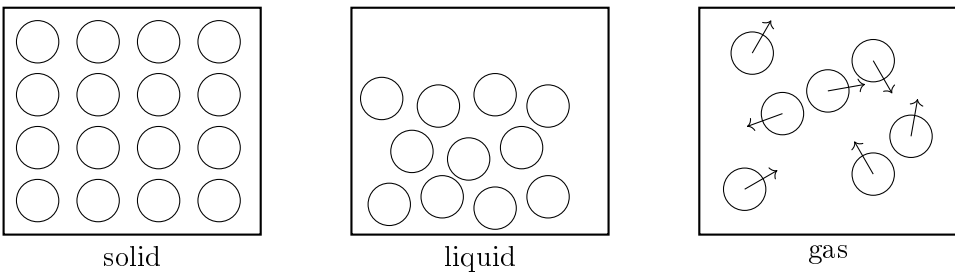


Figure 2: The same kind of particles in three states (two-dimensional model, schematic, not a real substance). Solid: regular pattern, vibrating in place. Liquid: close together but irregular, sliding past each other. Gas: far apart, fast, random motion (the arrows are the velocities).

Before moving on. What do you expect if I raise the temperature of a solid a little? And if I raise it more?

In the solid the particles sit in a regular pattern and vibrate. Raising the temperature they vibrate more, until the pattern breaks: the particles stay close but the pattern disappears,

and that is the liquid. Higher still, the particles leave each other: that is the gas, fast and with the particles far apart. On cooling, the particles do not “shrink”: the average spacing and the freedom to move change. The model includes walls and a small gravity only to keep the particles in the box. In the real world melting and boiling are also affected by pressure, which we ignore here.

The sealed syringe

(slide 8)

Before moving on. A sealed syringe is full of air. We push the plunger in until the volume is halved. The temperature does not change. What happens to the pressure of the gas? A: it decreases. B: it stays the same. C: it increases.

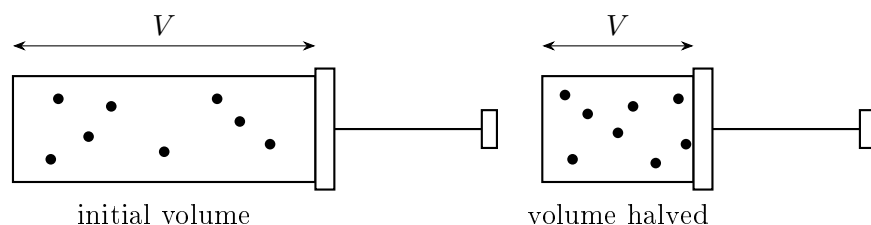


Figure 3: A sealed syringe full of air: the plunger is pushed in until the volume is halved, at the same temperature. The same particles in half the space hit the walls more often (schematic).

The answer is C: it increases. The same number of particles is squeezed into half the space: they hit the walls more often, so the pressure increases. The answer should contain “collisions”, “more often”, “the same particles”. We stay qualitative: the 0653 does not ask for $pV = \text{constant}$ here.

Gas pressure: collisions with the walls

(slide 9)

In the simulation, 80 non-interacting particles in two dimensions: the counter counts wall collisions per second and the pressure bar is computed from the momentum given to the walls (averaged over about 3 s, so it takes a few seconds to settle).

Lowering the piston to 50 %, the wall collisions per second go up and the pressure rises: the syringe prediction is confirmed.

Before moving on. If I heat the gas at constant volume, what changes in the collisions?

The particles move faster, so they hit the walls harder *and* more often. Here the temperature is shown in °C, because the kelvin comes in the next section.

While the piston moves, the temperature does *not* change in this model: the compression is slow and the walls conduct heat to the surroundings, so the gas stays at the same temperature (an isothermal compression). A fast compression would heat the gas, because the moving piston

gives kinetic energy to the particles that bounce on it: this is the effect behind the bicycle pump getting warm, which we keep for the lesson of 9 October. So here we isolate one cause at a time: the volume changes with T fixed, then T changes with the volume fixed.

Pressure is a statistical effect of a huge number of impacts. What is the force on the wall when only one particle hits it? Tiny and intermittent; but with 10^{23} particles it becomes steady.

What is temperature?

(slide 11)

The temperature of a substance is related to the average kinetic energy of its particles.

Before moving on. If I double the temperature in $^{\circ}\text{C}$, from 25°C to 50°C , does the average kinetic energy double?

No: in the simulation the bar reads $\times 1.08$. The bar is computed with the temperature in kelvin: $(50 + 273)/(25 + 273) = 323/298 \approx 1.08$. This prepares the next slide: we need a scale that starts at zero. Notice also that the particles do not all move at the same speed (the arrows have different lengths): we come back to this at slide 14. At -100°C the particles are slower, but not “frozen”: they are still moving.

Quick revision, and why we need it now

(slide 12)

$$T/\text{K} = \theta/^{\circ}\text{C} + 273$$

Given	Calculation	Result
25°C	$25 + 273$	298 K
77 K (liquid nitrogen boils)	$77 - 273$	-196°C
0 K	$0 - 273$	-273°C (absolute zero)
water boils, 100°C	$100 + 273$	373 K

Why now? The average kinetic energy is proportional to T in kelvin, not to θ in $^{\circ}\text{C}$. From 50°C to 100°C the energy is not doubled: $323 \rightarrow 373$ is only $\times 1.15$. Halving or doubling a temperature only makes sense on a scale that starts at the lowest possible temperature, where the particles have the minimum kinetic energy. A change of 1 K is the same as a change of 1°C : only the zero changes.

Three common mistakes

(slide 13)

Before moving on. What is wrong in each of these sentences? (1) “When a substance is heated, its particles get bigger.” (2) “Temperature and heat are the same thing.” (3) “At 0 K the particles disappear.”

1. The particles keep their size. They move faster (or vibrate more), and the spacing may change.
2. Temperature is related to the average kinetic energy of the particles. The thermal energy transferred also depends on how many particles there are.
3. They do not disappear: absolute zero is the lowest temperature, where the particles have the minimum possible energy.

“Average” does not mean “equal”

(slide 14)

At one temperature, the particles have a range of speeds. Temperature is linked to the *average* kinetic energy only. A few particles are much faster than the average; a few are much slower. Two particles in the same gas can have very different speeds.

Before moving on. If some particles are much faster than the average, what can they do that the average ones cannot?

We leave the question open: it prepares the puddle.

How are the speeds distributed?

(slide 15)

Beyond the 0653 syllabus: a support slide, only to understand the hook.

The horizontal axis is speed; the vertical axis is the fraction of particles per unit speed (figure 4). Three characteristic speeds, for nitrogen at 300 K: the most probable $v_p \approx 422$ m/s, the mean $\bar{v} \approx 476$ m/s, the root mean square $v_{\text{rms}} \approx 517$ m/s. They are in the ratio 1 : 1.128 : 1.225, always in this order, because the curve is asymmetric (long tail on the right). v_{rms} is the one linked to the mean kinetic energy: $\frac{1}{2}mv_{\text{rms}}^2 = \frac{3}{2}kT$ in three dimensions.

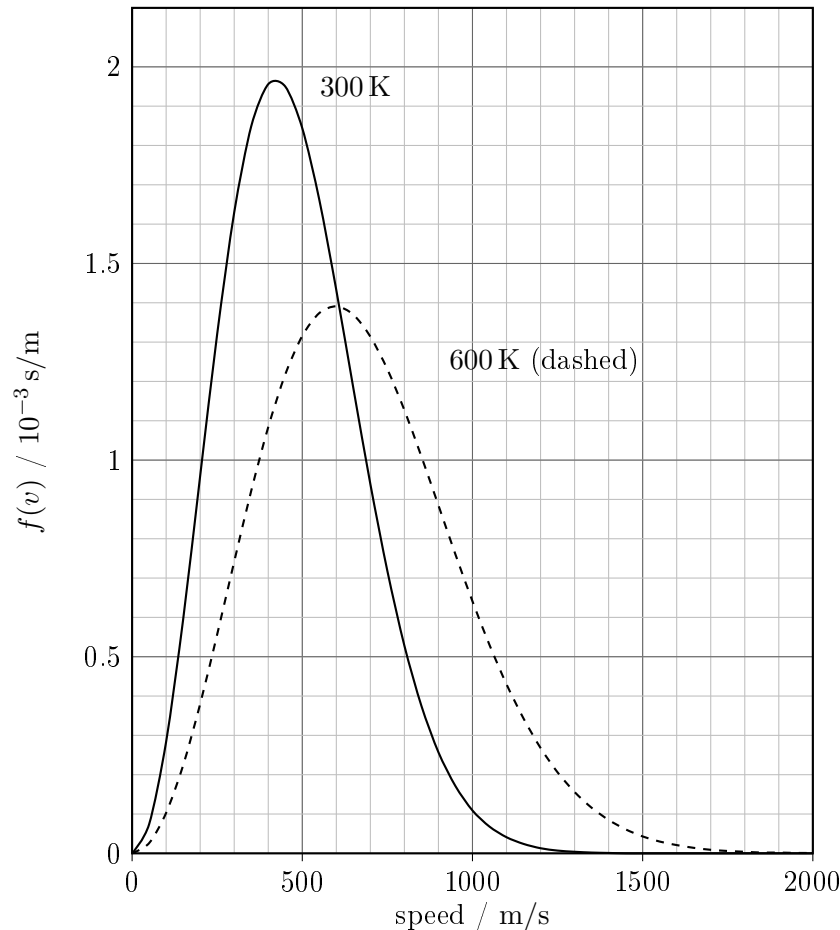


Figure 4: Maxwell-Boltzmann speed distribution for nitrogen (N₂) at 300 K (solid line) and at 600 K (dashed). The area under each curve is the same. Ideal gas; curves computed with $k_B = 1.380\,649 \times 10^{-23}$ J/K.

Raising the temperature, the peak moves to the right, gets lower and the curve widens (the area stays the same: the number of particles does not change). At a fixed speed $v^* = 1\,100$ m/s, the fraction of N₂ molecules faster than v^* goes from about 0.35 % at 300 K to about 7.9 % at 600 K. At the same temperature, lighter molecules are faster (same average kinetic energy, smaller mass).

Where does the shape come from?

(slide 16)

Beyond the 0653 syllabus: qualitative only, no derivation.

$$f(v) = 4\pi \left(\frac{m}{2\pi kT} \right)^{3/2} v^2 e^{-mv^2/2kT}$$

The factor v^2 rises with v : there are more ways to have a large speed. The exponential factor $e^{-E_k/kT}$ falls with v : a high kinetic energy $E_k = \frac{1}{2}mv^2$ is unlikely unless kT is large. The product has a maximum and a long tail. Raising T , kT gets larger and the exponential decays more slowly: the peak moves right (v_p proportional to $\sqrt{T}$), gets lower and the curve widens. The three characteristic speeds are all proportional to $\sqrt{kT/m}$:

$$v_p = \sqrt{\frac{2kT}{m}} \quad \bar{v} = \sqrt{\frac{8kT}{\pi m}} \quad v_{\text{rms}} = \sqrt{\frac{3kT}{m}}$$

(For N_2 , $m = 28.0134 \text{ u}$, $T = 300 \text{ K}$: 422/476/517 m/s.)

Back to the puddle

(slide 17)

The answer is B. At 20°C the average speed is not enough to break free. But some molecules at the surface are much faster than the average: they escape. The water **evaporates**. The molecules left behind are slower on average: the water **cools**. The key idea: average does not mean equal, the fastest particles can escape.

Idea A is the most natural. B is reconstructed with the fast tail of slide 15: at 20°C a few molecules are fast enough, and they are many in absolute number, because a drop contains billions of billions of molecules.

Back to the beakers: the ink in the hot water has spread more. The same mechanism: higher average speed, more collisions that carry the ink around (diffusion is faster at higher temperature). Why does a puddle dry faster in the sun? Higher temperature: a bigger fraction of fast molecules. And the wind? It removes the molecules above the surface so they do not come back: we leave it as a question for homework.

Evaporation and boiling

(slide 18)

	Evaporation	Boiling
Where?	only at the surface	throughout the liquid (bubbles of vapour)
Temperature?	at any temperature below the boiling point	only at the boiling point
Which particles?	the fastest ones escape	enough energy is supplied for particles throughout the liquid to form vapour
Effect on the liquid	the liquid cools	temperature stays constant while boiling (energy is supplied)
Speed	slow; faster if warmer, with a larger surface, with wind	rapid

“Constant temperature” while boiling holds at fixed pressure. The energy at the plateau goes into separating the particles (latent heat), which we use in the exercise without the formula. The distribution argument (the fastest particles) is our extension of the 0653.

Reading a heating curve

(slide 19)

Before moving on. If I keep heating melting ice, what does the thermometer do?

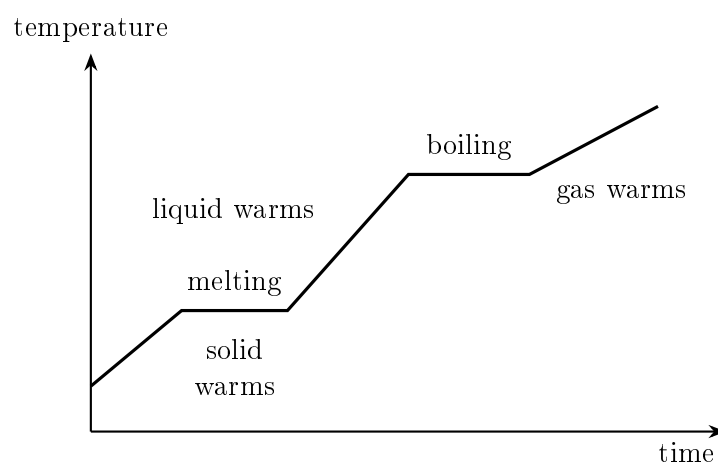


Figure 5: Heating curve, schematic (no numbers): sloping parts and two plateaus.

Sloping part. Energy is supplied and the temperature rises: the average kinetic energy of the particles increases. The state does not change.

Flat part (plateau). Energy is still supplied but the temperature stays constant. The substance is changing state: the energy is used to overcome the forces between the particles, not to make them move faster. The temperature of a plateau is the **melting point** or the

boiling point. The energy supplied does not increase the average kinetic energy, but the potential energy associated with the forces between particles, because they are being pulled out of their arrangement. We do not name latent heat or give formulas: 0653 level, qualitative.

Heating curve of a solid

(slide 20)

The graph shows the temperature of a substance being heated, against time (figure 6).

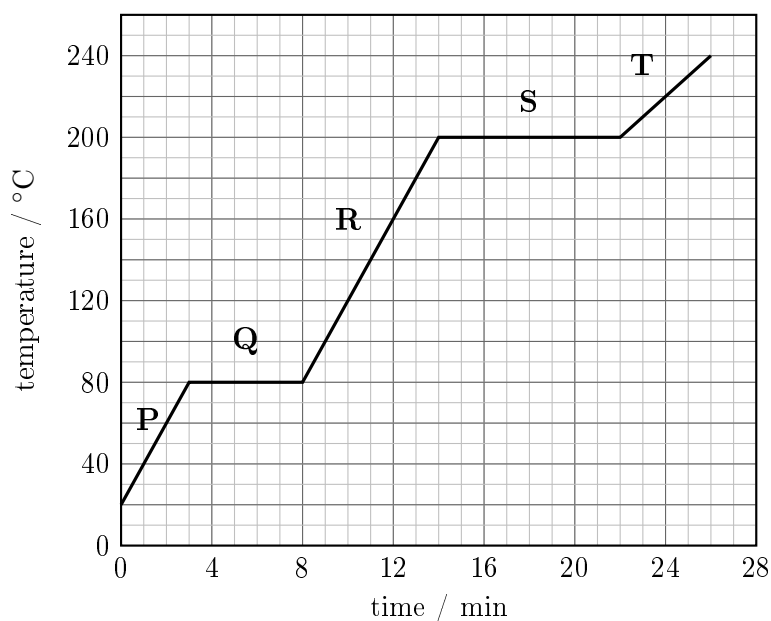


Figure 6: Heating curve of a solid: temperature against time. The regions are P, Q, R, S, T.

1. State the physical state of the substance in regions P, R and T.
2. Give the melting point and the boiling point.
3. In region Q, energy is supplied but the temperature does not rise. What happens to the energy?
4. Explain, using the particle model, why the temperature rises in region P.
5. In which region do the particles have the highest average speed?

The solutions are at the end of the chapter.

Homework

(slide 21)

To hand in at the next lesson, written answers: write complete sentences, not only words.

1. Explain, using the particle model, why a gas fills its container but a solid does not.
2. A student says: "When a gas is heated, its particles expand." Explain what is wrong and what really happens.
3. Explain why the pressure of the air in a sealed syringe rises when the plunger is pushed in.

4. Convert -40°C and 100°C to kelvin, and 250 K to $^{\circ}\text{C}$.
5. Explain why evaporation cools the liquid that remains.

Three points to keep

(slide 22)

1. Matter is made of particles that are always moving: Brownian motion is the evidence.
2. Gas pressure comes from collisions with the walls.
3. Temperature is related to the average kinetic energy of the particles, with T in kelvin.
4. Average does not mean equal: the fastest particles can escape, so water evaporates below 100°C .

Next lessons: on 9 October, heat and specific heat capacity.

Solutions to the exercises

(slide 20)

Heating curve (slide 20). Data of the curve: in P the temperature rises from 20°C to 80°C between 0 and 3 min; in Q it stays at 80°C until 8 min; in R it rises to 200°C between 8 and 14 min; in S it stays at 200°C until 22 min; in T it rises to 240°C at 26 min.

1. P: solid. R: liquid. T: gas.
2. Melting point 80°C , boiling point 200°C (the plateaus Q and S). They are temperatures, not times: 80°C , not 3 min.
3. In Q (melting) the energy is used to overcome the forces between the particles, which leave the fixed pattern. Their average kinetic energy does not change, so the temperature stays constant. The particles keep moving: the energy goes into the potential energy between the particles.
4. In P the energy supplied increases the average kinetic energy of the vibrating particles, so the temperature rises.
5. Region T: gas at the highest temperature, so the highest average kinetic energy (and average speed, for the same particles).

Homework (slide 21).

1. Gas: particles far apart, almost no forces, fast random motion, so they spread out to fill the space. Solid: strong forces hold the particles in fixed positions that they vibrate about.
2. The particles do not get bigger: they move faster, hit the walls harder and more often, so the pressure rises (or the volume increases, if the walls can move).
3. The same particles in a smaller volume: more collisions per second with the walls, so more pressure.
4. $-40 + 273 = 233\text{ K}$; $100 + 273 = 373\text{ K}$; $250 - 273 = -23^{\circ}\text{C}$ (watch the sign).
5. The fastest particles escape, so the average kinetic energy of the remaining particles decreases and the temperature falls.